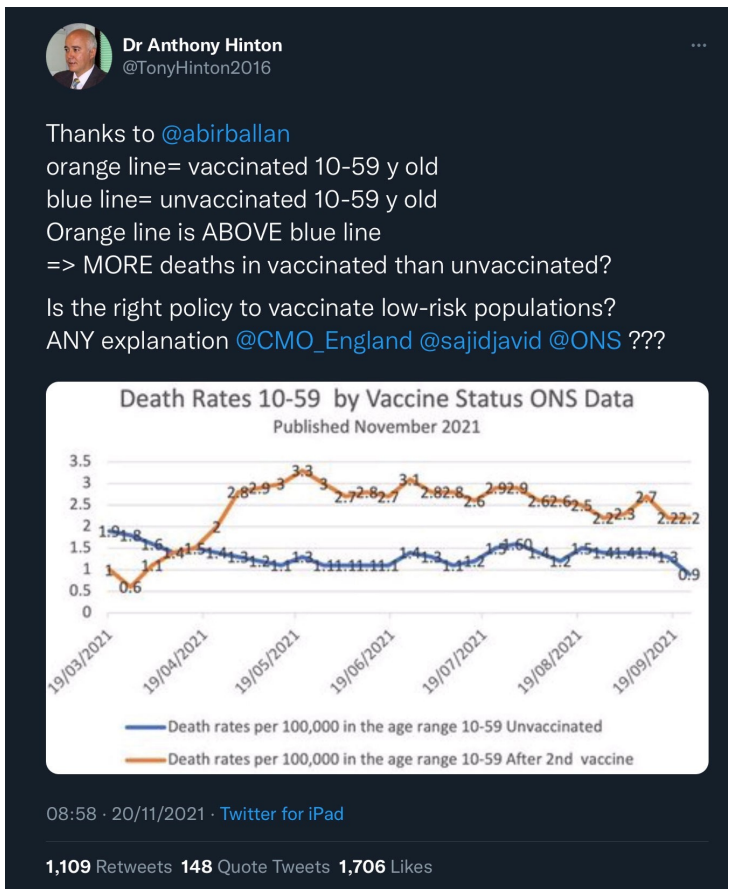
Case 2: Vaccination and deaths due to Covid

Comparing conditional distributions only makes sense if they are homogeneous with respect to other factors.



	Dead	NoDead	**Tot**
Vax	3512	281313	284825
NoVax	24364	2676016	2700380
Tot	27876	2957329	2985205

Conditional distributions by row

	Dead	NoDead	**Tot**
Vax	**0.0123**	0.9877	1
NoVax	**0.0090**	0.9910	1

Example: Consider the number of **covid-19** deaths among vaccinated and unvaccinated (age >= 18) in a given month (in the year 2021, at the beginning of the vaccination campaign)

The death rate among the vaccinated is higher than among the unvaccinated!

https://covidactuaries.org/2021/11/22/simpsons-paradox-and-vaccines/

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Frequency of death by vaccination status

	Dead	NoDead	Tot
Vax	**0.0123**	0.9877	1
NoVax	**0.0090**	0.9910	1

Number of deaths per 10000 people by vaccination status and age group

Age	Vax: Dead	Vax: Tot	(Dead|Vax)* 10000	NoVax: Dead	NoVax: Tot	(Dead|NoVax)* 10000
18-49	19	57154	3.32	541	1366914	3.96
50-59	29	31910	9.09	1225	513202	23.87
60-69	76	18981	40.04	2765	424476	65.14
70-79	381	57210	66.60	5901	303782	194.25
80-89	1760	100091	175.84	8679	73510	1180.66
90+	1247	19479	640.18	5253	18496	2840.07
Tot	3512	284825	**123.30**	24364	2700380	**90.22**

For each age group, the percentage of deaths is <u>lower</u> among the vaccinated than among the unvaccinated. However, when <u>age is not considered</u>, the relationship between the two percentages is <u>reversed</u>.

This is because: 1) the risk of dying from a covid increases with age 2) the percentage of unvaccinated people is much higher in the lower age groups (also due to decisions to prioritize the elderly in vaccination campaigns) and decreases significantly with age! Considering only aggregate data confuses the "effects" of vaccination and age.

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Simpson's paradox: When groups are **aggregated** (in this case, when age group is not taken into account), relationships between frequencies conditional on specific groups (in this case, relationships between death rates between vaccinated and unvaccinated by age group) may **disappear** or even **reverse**.

Comparing conditional distributions only makes sense if they are homogeneous with respect to possible confounding factors, which the researcher must necessarily identify and keep under control.